

3rd class

04.05.2026

problem: Find QD and CV of the following data:-

7, 8, 10, 12, 14, 13, 20, 25, 9.

→ Arranging the observation in ascending order,

7, 8, 9, 10, 12, 13, 14, 20, 25

Here, Total no. of observations,
 $n = 9$ which is odd No.

Therefore,

$$Q_3 = \frac{3(n+1)}{4} \text{ th obs.}$$

$$= \frac{3(9+1)}{4} \text{ " "}$$

$$= \frac{30}{4} \text{ " "}$$

$$= 7.5 \text{ th obs.}$$

$$= \frac{14+20}{2} = \frac{34}{2} = 17.$$

For Raw Data:

$$QD = \frac{Q_3 - Q_1}{2}, \text{CQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

Arranging the observation in ascending order.

When n is odd no.,

$$Q_3 = \frac{3(n+1)}{4} \text{ th observation}$$

$$Q_1 = \frac{n+1}{4} \text{ th " "}$$

When n is even no.,

$$Q_3 = \frac{\frac{3n}{4} \text{ th obs} + (\frac{3n}{4} + 1) \text{ th obs}}{2}$$

$$Q_1 = \frac{\frac{n}{4} \text{ th obs} + (\frac{n}{4} + 1) \text{ th obs}}{2}$$

$$Q_1 = \frac{n+1}{4} \text{th obs.}$$

$$= \frac{9+1}{4} \text{th obs.}$$

$$= 2.5 \text{th obs.}$$

$$= \frac{8+9}{2} \text{th observations}$$

$$= \frac{17}{2} \text{th obs.}$$

$$= 8.5 \text{th obs.}$$

$$Q_D = \frac{Q_3 - Q_1}{2}$$

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

$$= \frac{200 \left(1 + \frac{1}{A} \right) + 200 \left(\frac{1}{A} \right)}{2}$$

Ex 7, 8, 10, 12, 14, 13, 20, 25, 9, 10, 12, 17

⇒ Arranging the data in ascending order.

7, 8, 9, 10, 10, 12, 12, 13, 14, 17, 20, 25

Here, total No. of obs. $n = 12$, which is even number.

$$\therefore Q_3 = \frac{\frac{3n}{4} \text{th obs} + \left(\frac{3n}{4} + 1\right) \text{th obs}}{2}$$

$$= \frac{\frac{3 \times 12}{4} \text{th obs} + \left(\frac{3 \times 12}{4} + 1\right) \text{th obs}}{2}$$

$$= \frac{9 \text{th obs} + 10 \text{th obs}}{2}$$

$$= \frac{14 + 17}{2} = 15.5$$

$$Q_1 = \frac{\frac{n}{4} \text{th obs} + \left(\frac{n}{4} + 1\right) \text{th obs}}{2}$$

$$= \frac{\frac{12}{4} \text{th obs} + \left(\frac{12}{4} + 1\right) \text{th obs}}{2}$$

$$= \frac{3 \text{th obs} + 4 \text{th obs}}{2} = \frac{9 + 10}{2} = 9.5.$$

$$\therefore QD = \frac{Q_3 - Q_1}{2} = \frac{15.5 - 9.5}{2} = 3$$

$$CQD = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

$$= \frac{6}{25} \times 100$$

$$= 24\%$$

problem : find QD and CV;

**

$$Q_1 = L_1 + \frac{\frac{N}{4} - F_{Q_1-1}}{f_{Q_1}} \times C_1$$

Where,

$N = \sum_{i=1}^n f_i$ = Summation of frequencies.

L_1 = Lower Limit of Q_1 class

f_{Q_1} = Frequency of Q_1 class

F_{Q_1-1} = Cumulative frequency of pre- Q_1 class.

** For Group Data,

$$Q_3 = L_3 + \frac{\frac{3N}{4} - F_{Q_3-1}}{f_{Q_3}} \times C_3$$

Where,

$N = \sum_{i=1}^n f_i$ = Summation of the frequencies.

L_3 = Lower Limit of Q_3 class

f_{Q_3} = frequency of Q_3 class

C_3 = Length or size of Q_3 class

F_{Q_3-1} = Cumulative frequency of pre- Q_3 class.

problem: Find QD and CQD

C1	Frequency (f_i)
10-20	2
20-30	8
30-40	10
40-50	12
50-60	6

Note:

Q_3, Q_1 বের করার জন্য Cumulative freq. বের করতে হবে।

⇒

C1	Frequency (f_i)	Cumulative Freq (F_i)
10-20	2	2
20-30	8	10
30-40	10	20
40-50	12	32
50-60	6	38

$$N = 38$$

$$Q_1 \text{ Class} = \frac{N}{4}$$

$$= \frac{38}{4} = 9.5$$

9.5 Cumulat. 10 এর অর্ধেক হবে তাই এটা Q_1 Class

Q_3 Class

$$Q_3 \text{ class} = \frac{3N}{4}$$

$$= \frac{3 \times 38}{4} = 28.5$$

Here, $L_1 = 20$, $f_{Q_1} = 8$

$$F_{Q_1-1} = 2, C_1 = 10$$

$$L_3 = 40, f_{Q_3} = 12$$

$$F_{Q_1-1} = 20, C_3 = 10$$

We know,

$$Q_3 = L_3 + \frac{\frac{3N}{4} - F_{Q_3-1}}{f_{Q_3}} \times C_3$$

$$= 40 + \frac{28.5 - 20}{12} \times 10$$

$$= 47.083$$

$$Q_1 = L_1 + \frac{\frac{N}{4} - F_{Q_1-1}}{f_{Q_1}} \times C_1$$

$$= 20 + \frac{9.5 - 2}{8} \times 10$$

$$= 29.375$$

(i) Frequency	(ii) Cumulative Frequency
5	5
8	13
10	23
12	35
3	38

(i) Frequency	(ii) Cumulative Frequency
5	5
8	13
10	23
12	35
3	38

$$\begin{aligned}
 QD &= \frac{Q_3 - Q_1}{2} \\
 &= \frac{47.083 - 29.375}{2} \\
 &= 8.854
 \end{aligned}$$

$$\begin{aligned}
 CQD &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100 \\
 &= \frac{47.083 - 29.375}{47.083 + 29.375} \times 100 \\
 &= 23.16 \%
 \end{aligned}$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$-1 \leq r \leq 1$$

$$\frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n^2} - \bar{x} \bar{y}$$

$$r = \frac{\left\{ \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 \right\} \left\{ \frac{\sum_{i=1}^n y_i^2}{n} - \bar{y}^2 \right\}}{\sqrt{\left\{ \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 \right\} \left\{ \frac{\sum_{i=1}^n y_i^2}{n} - \bar{y}^2 \right\}}}$$

05-05-2026

4th Class

Correlation

Correlation measures the degree of relationship between two variables. Let, x and y be two quantitative variables, then the correlation between x and y is measured by the Karl's Pearson's correlation coefficient.

It is usually denoted by r or r_{xy} . It can be formulated

$$r_{xy} = r = \frac{\sum_{i=1}^n (\bar{x}_i - \bar{x})(\bar{y}_i - \bar{y})}{\sqrt{\sum_{i=1}^n (\bar{x}_i - \bar{x})^2 \sum_{i=1}^n (\bar{y}_i - \bar{y})^2}}$$

r must lie in $-1 \leq r \leq 1$.

$$r = \frac{\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n}}{\sqrt{\left\{ \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right\} \left\{ \sum_{i=1}^n y_i^2 - \frac{\left(\sum_{i=1}^n y_i \right)^2}{n} \right\}}}$$

It is also called correlation coefficient.

problem : Draw Box plot and find the outliers of the following data -

weight (in kg.) of new baby born:

1.5, 2, 2.5, 3, 3.5, 4, 5, 10, 3

Solⁿ:

Arranging the observations in ascending order.

1.5, 2, 2.5, 3, 3, 3.5, 4, 5, 10

Here, $N = 9$ which is odd.

$$\text{First Quartile } Q_1 = \frac{(n+1)}{4} \text{th obs.}$$

$$= \frac{9+1}{4} \text{ " "}$$

$$= \frac{10}{4} \text{ " "}$$

$$= 2.5 \text{ th obs.}$$

$$= \frac{2+2.5}{2} \text{ (average 2 वा 3वां)} \\ = 2.25$$

$$\text{Third Quartile, } Q_3 = \frac{3(n+1)}{4} \text{ th obs.}$$

$$= \frac{3(9+1)}{4} \text{ " "}$$

$$= \frac{3 \times 10}{4} \text{ " "}$$

$$= 7.5 \text{ th obs} = \frac{4+5}{2} = 4.5$$

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

$$= 4.5 - 2.25$$

$$= 2.25$$

$$\text{Median} = \frac{n+1}{2} \text{th obs.}$$

$$= \frac{9+1}{2} \text{th obs.}$$

$$= 5 \text{th obs.}$$

$$= 3$$

$$\text{Minimum limit} = Q_1 - 1.5 \times \text{IQR}$$

$$= 2.25 - 1.5 \times 2.25$$

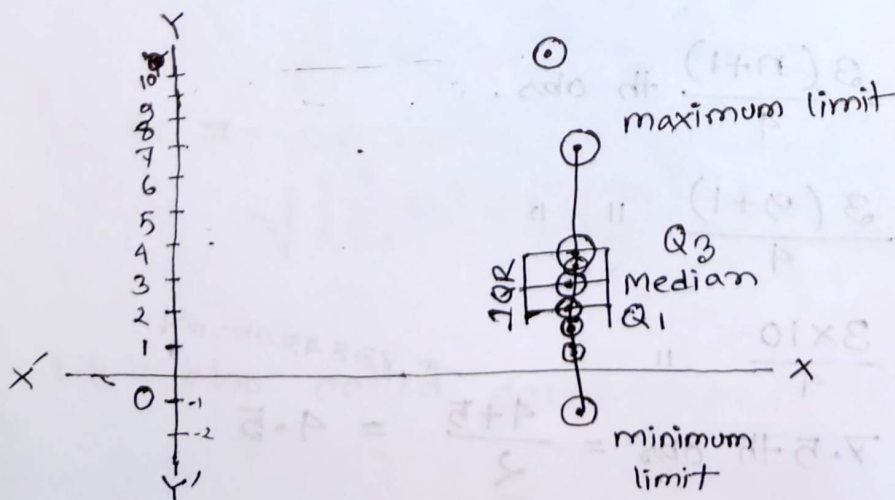
$$= -1.125$$

$$\text{Maximum limit} = Q_3 + 1.5 \times \text{IQR}$$

$$= 4.5 + 1.5 \times 2.25$$

$$= 7.875$$

limit
minimum এর চেয়ে ছোট
maximum " " বড়
(যেগুলো থাকবে যব
outlier)



1 small squ. = 1 unit X axis

5 small square = 1 unit in Y.
axis

(১ টা data ই বসাতে হবে)

Fig: Box-Plot

from the Box Plot, the observation, 10 kg is an outlier which exceeds the maximum limit.

problem : The following data represent the ages of husbands and the ages of wives of 7 couples as follows -

Age of Husbands	30	21	40	60	50	22	30
Age of wives	25	18	45	24	30	40	25

- i) Draw Scatter plot and comment on the relationship between the ages of husbands and the ages of wives.
- ii) Find the correlation coefficient and make comment about the relationship between the ages of husbands and the ages of wives.

Correlation is five types.

① positive correlation ($-1 < r < 0$) ($0 < r < 1$)

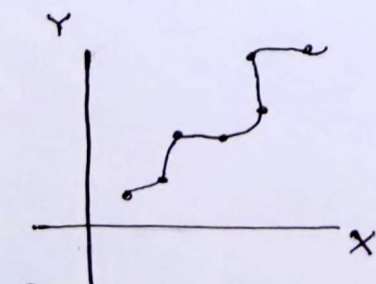


fig : Scatter Plot

উৎসাহী শ্রম positive

ii) perfectly positive correlation ($r = 1$)

x এর মান বাড়লে y এর মান 3 গুণবে।

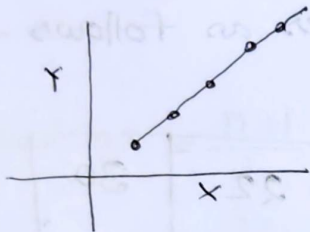


fig: Scatter plot

iii) Negative Correlation ($-1 < r < 0$)

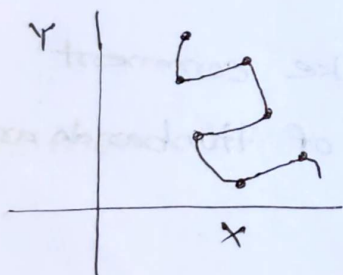


fig: Scatter plot

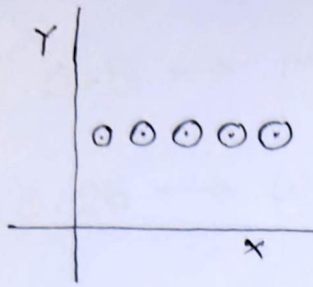
নিম্নস্বামী Husb/wife এর বয়স

iv) perfectly Negative correlation ($r = -1$)



fig: Scatter plot.

v) zero correlation



(x এর মান change হচ্ছে but y same থাকছে
সম্পর্ক নাই)

fig: Scatter plot

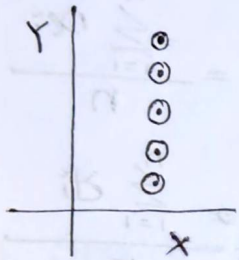
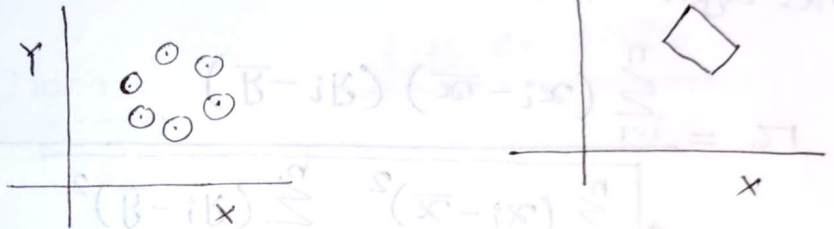


fig: Scatter plot



(কোনো trend follow না করলেই zero correlation)
(সবগুলোই zero correlation)

1) Solⁿ :- Husband & wife's age এর মধ্যে correlation :

11) Let,

x_i = The ages of Husbands in years.

y_i = " " " Wives " "

∴ The correlation coefficient between the ages of husbands & the ages of wives is -

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

Necessary calculating table

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
$\sum_{i=1}^n x_i$	$\sum_{i=1}^n y_i$			$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$	$\sum_{i=1}^n (x_i - \bar{x})^2$	$\sum_{i=1}^n (y_i - \bar{y})^2$

$(\pm) 0.75 \rightarrow \text{Strongly } (\pm)$

$(\pm) 0.5 \rightarrow \text{Moderate}$

$(\pm) 0.25 \rightarrow \text{Weak.}$

যদি প্রকৃতির বন্যজাতি গাছ
আমের ১৫ প্রকার
Strong / Moderate / Weak হবে।

6th class

12.05.2026

▣ Spear Man's Rank Correlation:

The correlation between two qualitative variables is measured by the Spear Man's Rank correlation coefficient.

It is usually denoted by R (Row) and it can be defined

as,

$$R = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

where, d_i = The difference between the ranks of two qualitative variables.

n = No. of Samples.

problem : The following data represent the marks given by Judge A and Judge B in a Fashion Show competition as follows -

SN	Marks given by Judge A	Marks given by Judge B
1	60	55
2	65	50
3	70	75
4	60	55
5	80	75
6	60	52
7	60	50

- i) Draw Scatter plot and find the correlation between the marks given by Judge A and Judge B.
- ii) Find the correlation coefficient between the marks given by Judge A and Judge B and make comment about the relationship.

Solⁿ:

(ii) Let,

x_i = The marks given by Judge A

R_{xi} = The Ranks of the marks given by Judge A.

d_i = The marks given by judge B.

R_{yi} = The Ranks of the marks given by Judge B.

We know,

Correlation Coefficient,
$$r = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

$$d_i = R_{xi} - R_{yi}$$

$$n = 7.$$

Necessary Calculating Table :-

SN	Marks given by Judge A	Marks given by Judge B	R_{xi}	R_{yi}	$d_i = R_{xi} - R_{yi}$	d_i^2
1	60	55	5.5	3.5	2	
2	65	50	3	6.5	-3.5	
3	70	75	2	1.5	.5	
4	60	55	5.5	3.5	2	
5	80	75	1	1.5	-.5	
6	60	52	5.5	5	.5	
7	60	50	5.5	6.5	-1	

আগে highest v. সূচকের 1 টা করে মান থাকার ক্ষেত্রে
60 পাঠে 4 1, 5, 6, 7 = $\frac{22}{4} = 5.5$ - তবে
এস → position

$$\frac{1+2}{2} = 1.5, \frac{3+4}{2} = 3.5$$

$$\sum_{i=1}^n d_i^2 =$$

$$\therefore r = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 22}{7(49-1)} = 1 - \frac{132}{7 \times 48}$$

$$= 0.607$$

Comment : The correlation coefficient, $r = 0.607$ which is approximately near to 0.5, Therefore, there exist approximately ^{Moderate} ~~Strong~~ position correlation between the marks given by Judge A and Judge B. It is approximately ^{Moderately} ~~Strong~~ evident that the judges were unbiased.

* negative আসলে biased.
positive " unbiased.

Regression :-

Francis Galton 'regression' আবিষ্কার করেন।

⇒ Regression means Stepping back towards the average of the dependent variables with respect to the corresponding changes of independent variables.

Let, us consider two variables linear regression model as

$$y = \alpha + \beta x + e \quad (y = mx + c \text{ এর মতো})$$

where,

y = dependent variable

x = independent variable

α = Intercept

β = Regression coefficient.

e = Error / residual / disturbance / noise

Intercept of y	Intercept of x
0.1	0.1
0.2	0.2
0.3	0.3
0.4	0.4
0.5	0.5

OLSE use করে
error কে minimize
করা হয়।

∴ The estimated Model is

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

where

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

$$\text{and } \hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

7th Class

17.05.2026

problem: The following data represent the income and expenditure of 5 respondent as follows-

Income in Thousant Taka	Exp. in thousant Tk
20	19
24	18
30	31
50	48
25	20

ind $\rightarrow x$ (Income)

dep $\rightarrow y$ (Expenditure)

- i) Estimate the linear regression model of expenditure on income.
- ii) Predict the expenditure if income is 35 Thousand Tk.

Solⁿ:

Let, x_i = The income of the respondent in thousand

y_i = " expenditure of " " " " "

$\therefore$ The estimated regression model of y on x is,

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

where,

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \quad \&$$

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n}}{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}}$$

∴ Necessary Calculating Table

x_i	y_i	x_i^2	$x_i y_i$
20	19		
24	18		
30	31		
50	48		
25	20		

$\sum_{i=1}^n x_i$	$\sum_{i=1}^n y_i$	$\sum_{i=1}^n x_i^2$	$\sum_{i=1}^n x_i y_i$
= 149	= 136	= 5001	= 4642

$$\therefore \hat{\beta} = \frac{4642 - \frac{149 \times 136}{5}}{5001 - \frac{(149)^2}{5}}$$

$$\therefore \hat{\beta} = 1.0506$$

যদি (i) জান থাকতাহ (এ)
খরচ বাড়েনি
যদি (ii) জান থাকতাহ
তাহ (এ খরচ তোড়ে

$$\therefore \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

$$= \frac{\sum_{i=1}^n y_i}{n} - \hat{\beta} \frac{\sum_{i=1}^n x_i}{n}$$

$$= \frac{136}{5} - 1.0506 \times \frac{149}{5}$$

$$= -4.109$$

মান যদি,

(+) income এর সাথে উল্লভ হবে

(-) income এর সাথে উল্লভ হবে

— মাঝে exp 4 শতাব্দীর জন্য হবে।

i) The estimated model is

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

$$\hat{y} = -4.109 + 1.0506 x \quad \text{Ans.}$$

ii) If $x = 35$ then predicted expenditure will be

$$\hat{y}_{35} = -4.109 + 1.0506 \times 35$$

$$= 32.662$$

thousand Tk.

** probability থেকে $2/3$ টা set আছে আর বাকীগুলো থেকে 2 set.

* correlation থেকে - Mandatory.

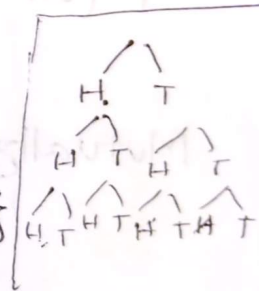
probability

Define: (i) Experiment (ii) Trial (iii) Outcomes (iv) Sample Space

(v) Event (vi) probability of an Event

Trial : কতবার টস করে ফল পলায় (যেই তার Trial, যদি 3 বার টস করা পর outcome আয় তাহলে Trial - 3.

Outcomes এর set $\Rightarrow$ Sample Space : $S = \{H, T\}$



$$S = \{HH, HT, TH, TT\}$$

3rd time toss

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

যদি বনে, $S = \{S, F\}$

1 বার toss

$$\therefore S = \{SS, Sf, FS, FF\}$$

2 বার toss

one or more outcomes ~~for~~ $\rightarrow$ ~~the~~ event.

probability of an Event:

$$P(A) = \frac{\text{Favourable no. of outcomes of Event A}}{\text{Total no. of outcomes}}$$

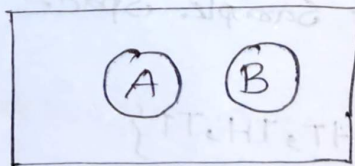
$$P(A) = \frac{m}{n}$$

$$0 \leq P(A) \leq 1$$

Tossing a coin 3 time

$$P(\text{at least one Head}) = \frac{7}{8}$$

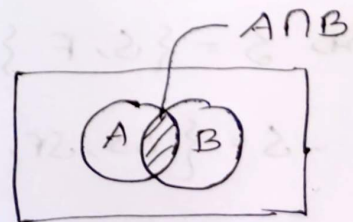
Mutually Exclusive Event :-



$$P(A \cup B) = P(A) + P(B)$$

Mutually inclusive Event :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



A & B are called independent.

$$P(A \cap B) = P(A) \cdot P(B)$$

Conditional probability :-

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

problems :-

The following data represent the status as follows -

Having Status	Having House	Not Having House	Total
Having Car	40	70	110
Not Having Car	60	80	140
Total	100	150	250

If a respondent is randomly selected then

find the following probabilities -

i) The respondent is having House

ii) " " " " Car

iii) " " " " House & Car

iv) " " " " Neither House nor Car

v) " " " " House or Car.

vi) House when having car is given.

Solⁿ:-

Let, Having House = H

Not " " = H'

Having Car = C

Not " " = C'

$$i) P(H) = \frac{100}{250} =$$

$$ii) P(C) = \frac{110}{250} =$$

$$iii) P(H \cap C) = \frac{40}{250} =$$

$$iv) P(H' \cap C') = \frac{80}{250} =$$

$$v) P(H \cup C) = P(H) + P(C) - P(H \cap C)$$

$$vi) P(H/C) = \frac{P(H \cap C)}{P(C)}$$

#	x	0	1	2	3
	$P(x=x)$	0.5	0.4	0.2	-0.1

$$\textcircled{1} 0 \leq P(x) \leq 1$$

$$\textcircled{2} \sum P(x) = 1 \text{ হওয়া}$$

Q জ্ঞা.

= Make comment it is right or

wrong? $\rightarrow$ condition 2 টা সত্যি করে না তাই

This is not valid

#	x	0	1	2	3
	$P(x=x)$	0.5	0.2	0.1	0.2

এই probability
valid or not?

#	No. of System Sold	Gain/Loss (in Tk)	probability $P(y=y)$
	0	0	0.5
	1	5000	0.25
	2	10000	0.15
	3	15000	0.10

find the expected sales.

Let,

$\Rightarrow$ Sale = y

$$\therefore E(y) = \sum y P(y)$$

যদি exp. x বিধি,

$$E(x) = \sum x P(x)$$

$$= 0 \times 0.5 + 5000 \times 0.25 + 10000 \times 0.15 + 15000 \times 0.10$$

=

correlation & regression
CT रान

Discrete	continuous
$P(x)$ ↓ pmf probability Mass -func	$f(x)$ ↓ pdf probability density -func
* $E(x) = \sum x P(x)$	* $E(x) = \int x f(x) dx$

$f(x) = x^2 + 3x + 2$

$; 0 < x < 7$

Q. Is this discrete or continuous?

⇒ It is continuous.

$$E(x) = \int_0^7 x f(x) dx$$

$$= \int_0^7 (x^2 + 3x + 2) dx$$

$$= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_0^7$$

(Am)